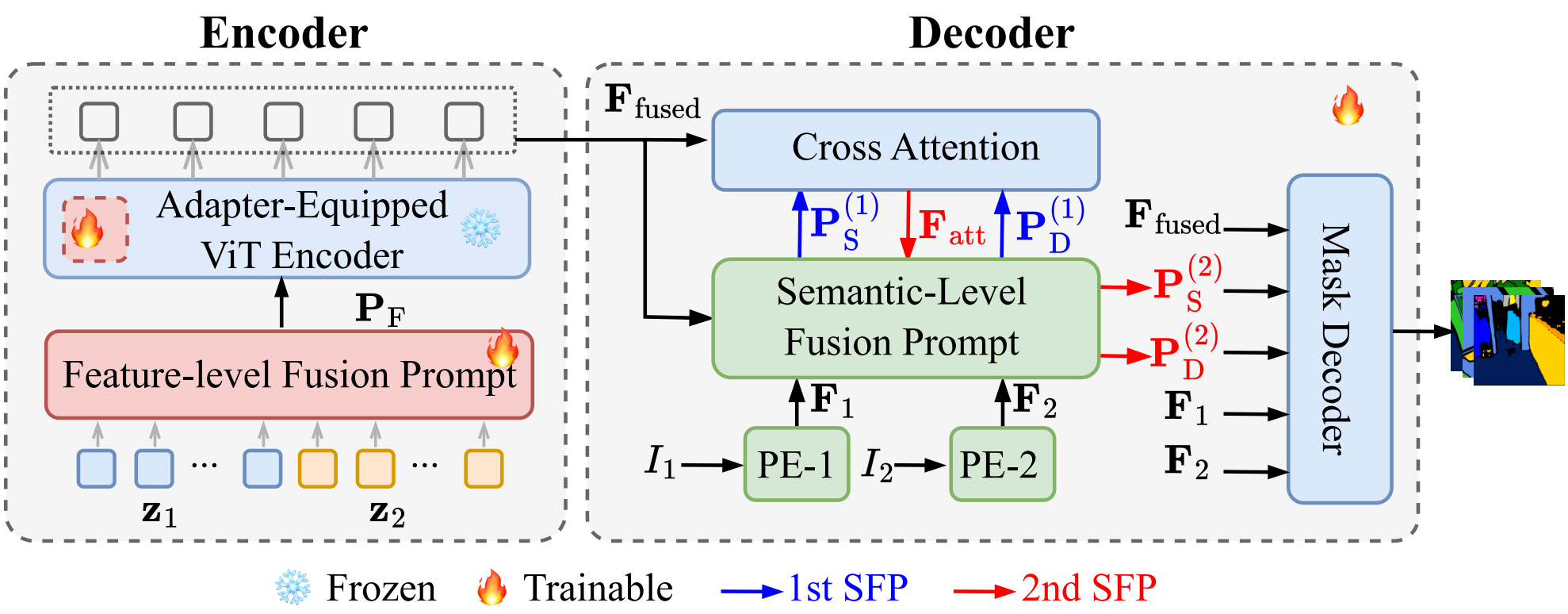
Encoder
Decoder
Adapter-Equipped ViT Encoder
$\mathbf{P}_F$
Feature-level Fusion Prompt
$\mathbf{z}_1$
$\mathbf{z}_2$
$\mathbf{F}_{\text{fused}}$
Cross Attention
$\mathbf{P}_S^{(1)}$
$\mathbf{F}_{\text{att}}$
$\mathbf{P}_D^{(1)}$
Semantic-Level Fusion Prompt
$\mathbf{F}_1$
$\mathbf{F}_2$
I_1
PE-1
I_2
PE-2
$\mathbf{F}_{\text{fused}}$
$\mathbf{P}_S^{(2)}$
$\mathbf{P}_D^{(2)}$
$\mathbf{F}_1$
$\mathbf{F}_2$
Mask Decoder
Frozen
Trainable
1st SFP
2nd SFP